Cycle Test 1: worked solutions

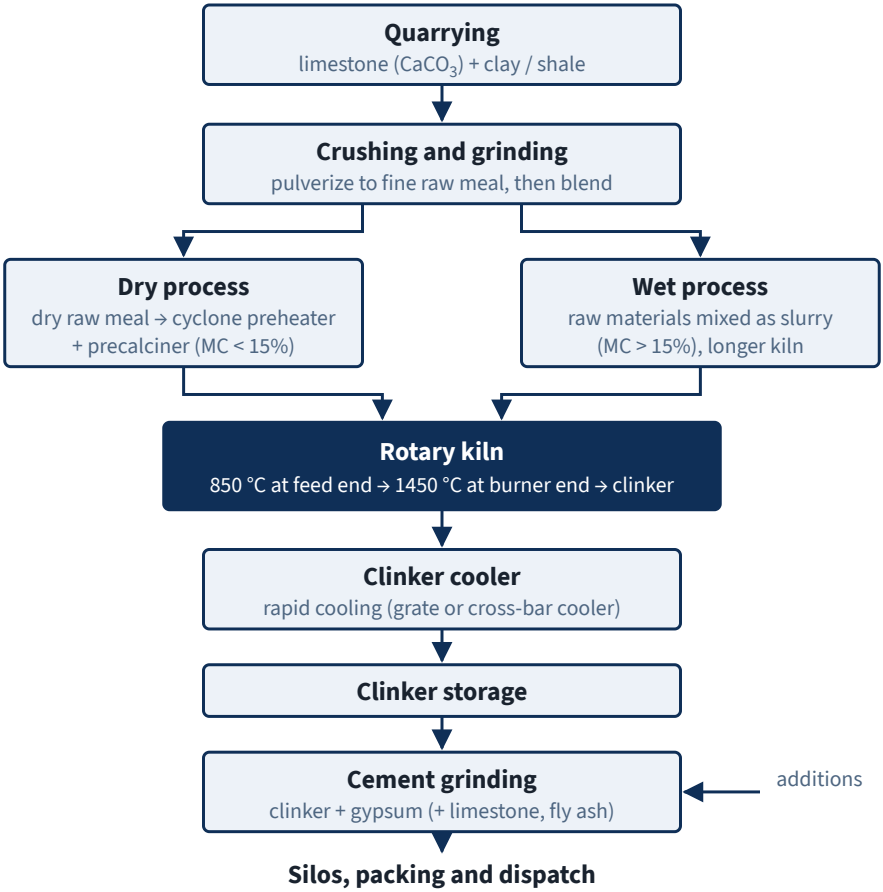
19 Aug 2025 · 30 marks · 1 hour · CEPC14

Part A

Q1

1 mark

> Explain the process of making cement in a flowchart.



Quarrying of limestone and clay → crushing and grinding into raw meal → blending (dry or wet) → preheater/precalciner → **rotary kiln (up to 1450 °C) → clinker** → rapid cooling → **grinding with about 3–5% gypsum** → storage, packing and dispatch.

Q2

1 mark

> Which among the constituents of ordinary Portland cement composition is wrong?
> a) CaO 60–67% b) SiO_2 17–25% c) Al_2O_3 8.0–15.0% d) Fe_2O_3 0.5–6.0%

(c) Al_2O_3 8.0–15.0% is wrong. The usual range of alumina in OPC is **3–8%**.

Q3

1 mark

> Write the chemical name of Bogue compounds.

Short form	Chemical name	Formula
C_3S	**Tricalcium silicate** (alite)	$3CaO{\cdot}SiO_2$
C_2S	**Dicalcium silicate** (belite)	$2CaO{\cdot}SiO_2$
C_3A	**Tricalcium aluminate**	$3CaO{\cdot}Al_2O_3$
C_4AF	**Tetracalcium aluminoferrite**	$4CaO{\cdot}Al_2O_3{\cdot}Fe_2O_3$

Q4

1 mark

> In Bogue's composition, which compound has the highest heat of hydration? a) C_3S b) C_2S c) C_3A d) C_4AF

(c) C_3A, about **867 J/g** (C_3S 502, C_4AF 419, C_2S 260 J/g).

Q5

1 mark

> Which Bogue's compound is least reactive in the hydration process?

(b) C_2S. It reacts slowly and gives later-age strength with the lowest heat (260 J/g).

Q6

1 mark

> What type of cement is typically used for mass concrete works such as dams, why?

Low heat Portland cement (IS 12600; ASTM Type IV), or blended cements such as PPC/PSC. It has **less C_3S and C_3A and more C_2S**, so it releases **heat slowly**. In a dam the heat cannot escape from the core; a fast heat release would cause a large temperature difference between core and surface, and **thermal cracking**.

Q7

1 mark

> The required concrete grade is M80. What type of aggregate will you choose in terms of aggregate impact value? a) 30% b) 20 c) 40% d) 50%

(b) 20%. A lower impact value means a **tougher** aggregate. For high-strength concrete the aggregate itself can become the weak link, so choose the strongest, toughest aggregate (well below the 30% limit for wearing surfaces and 45% for other concrete).

Q8

1 mark

> The needle-like crystals in the hydrated cement paste formed in the early hydration stage are called _____.

Ettringite (AF_t, $C_6A\bar{S}_3H_{32}$), formed from C_3A + gypsum.

Q9

1 mark

> Sulphate resisting cement will have: a) Low C_3A and high C_4AF b) High C_3A and high C_4AF c) Low C_3A and low C_4AF d) High C_3A and low C_4AF

(a) Low C_3A and high C_4AF. C_3A is the compound that reacts with sulfates to form expansive ettringite, so it is kept low (IS 12330: $C_3A \leq 5\%$; $2C_3A + C_4AF \leq 25\%$). The C_4AF content is relatively higher (ASTM Type V: C_3A 0–4%, C_4AF 10–12%).

Q10

1 mark

> Limit for chlorides and sulphites content in water used for concrete mixing is ___ ppm and ___ ppm respectively.

As taught in the lectures: **chlorides < 500 ppm** and **sulfates (as SO_3) < 1000 ppm**.

(IS 456:2000 Cl. 5.4 gives: chlorides 2000 mg/l for plain concrete and **500 mg/l for reinforced concrete**; sulfates as SO_3 **400 mg/l**.)

Q11

2 marks

> What is meant by hydration? Write the chemical equations when compounds of cement react with water.

Hydration is the **exothermic chemical reaction between cement and water** that forms hydration products (mainly C-S-H gel and $Ca(OH)_2$), causing setting and hardening.

$$2C_3S + 6H \rightarrow C_3S_2H_3 + 3CH$$

$$2C_2S + 4H \rightarrow C_3S_2H_3 + CH$$

$$C_3A + 3C\bar{S}H_2 + 26H \rightarrow C_6A\bar{S}_3H_{32} \text{ (ettringite)}$$

$$2C_3A + C_6A\bar{S}_3H_{32} + 4H \rightarrow 3C_4A\bar{S}H_{12} \text{ (monosulphate)}$$

$$2C_3A + 21H \rightarrow C_4AH_{13} + C_2AH_8 \text{ (without gypsum: flash set)}$$

(C = CaO, S = SiO_2, A = Al_2O_3, H = H_2O, $\bar{S}$ = SO_3, CH = $Ca(OH)_2$.)

Q12

2 marks

> Building needs to be built in a cold region (temperature below 1 °C). Which admixtures will you choose and why?

- **Accelerator** (non-chloride, e.g. **calcium nitrate, calcium formate, triethanolamine**): speeds up setting and early strength so the concrete gains strength (about 3.5–5 MPa) before it can freeze, and allows earlier formwork removal. **Avoid $CaCl_2$ in reinforced concrete** because chlorides cause corrosion.
- **Air-entraining admixture**: tiny air bubbles relieve the pressure when pore water freezes, giving **freeze–thaw resistance**.
- **Antifreeze admixture** (e.g. calcium nitrite/nitrate, glycols): lowers the freezing point of the mixing water.

(Along with heated materials and insulated formwork.)

Q13

2 marks

> How will you check the soundness of cement? What makes the cement unsound?

Unsoundness is excessive expansion of hardened cement caused by delayed hydration of **free lime (CaO)**, **magnesia (MgO)** and **excess gypsum** (late ettringite).

Tests (IS 4031 Part 3):

- **Le Chatelier test** (free lime only): paste of standard consistency in the Le Chatelier mould, kept in water 24 h, then boiled; the increase in distance between the indicator points must not exceed **10 mm**.
- **Autoclave test** (free lime + MgO): neat cement bar steam-cured at **2 MPa** for 3 h; expansion must be **less than 0.8%**.

Part B

Q14

2 marks

> Select the size of aggregate for a concrete mix for a slab thickness of 50 mm with a spacing of reinforcement of 50 mm. What will be the limiting value of impact resistance for the aggregate?

Maximum size of aggregate:

- ≤ 1/4 of the minimum thickness (IS 456) → 50/4 = 12.5 mm (or ≤ 1/5 of the narrowest form dimension → 10 mm);
- ≤ clear spacing between bars − 5 mm → 45 mm (or ≤ 3/4 of clear spacing → 37.5 mm).

The thickness governs: **use 10 mm nominal maximum size aggregate**.

Limiting aggregate impact value (IS 383): **30%** if the slab is a **wearing surface** (floor, pavement); **45%** for other concrete.

Q15

2 marks

> What is the need for mineral admixtures other than reduction of cement consumption? Generally mineral admixtures react with what in case of ordinary Portland cement?

Needs (besides saving cement):

- **Durability**: convert the weak, soluble **$Ca(OH)_2$** into extra **C-S-H**; **pore refinement** and denser ITZ → lower permeability; better resistance to **sulfates, chlorides and ASR**.
- **Lower heat of hydration** (mass concrete).
- **Higher long-term strength**; very high strength with silica fume.
- **Workability**: fly ash spheres act like ball bearings; less bleeding and segregation.
- **Sustainability**: uses industrial by-products, lowers CO_2.

They react with the **calcium hydroxide (CH)** released by OPC hydration (pozzolanic reaction): CH + S + H → C-S-H.

Q16

2 marks

> What is meant by lime saturation factor? What is the maximum and minimum limit for the cement?

LSF is the ratio of the lime actually present to the lime needed to combine fully with silica, alumina and iron oxide. It controls how much **free lime** remains in the clinker.

$$\text{LSF} = \frac{\text{CaO} - 0.7\ \text{SO}_3}{2.8\ \text{SiO}_2 + 1.2\ \text{Al}_2\text{O}_3 + 0.65\ \text{Fe}_2\text{O}_3}$$

Limits (IS 269:2015): **maximum 1.02**; **minimum 0.66** for OPC 33 and **0.80** for OPC 43/53. In practice clinker LSF is kept at **0.92–0.98**. LSF above 1.0 leaves free lime → unsoundness.

Q17

2 marks

> What is meant by bulking of sand? Will it happen in all types of aggregate?

Bulking is the **increase in volume of moist sand** compared with dry sand. Thin water films around the grains hold them apart (surface tension), so the sand packs loosely. Bulking rises with moisture up to about **4–6%** (volume increase **20–40%**), then falls; **fully saturated sand does not bulk**. Finer sand bulks more.

No, it does not happen to the same extent in all aggregates: it is significant only for **fine aggregate**. **Coarse aggregate bulks negligibly**, because the water films are tiny compared with the particle size. It matters when sand is batched **by volume**.

Q18

4 marks

> Two samples of sieve analysis results are given below. Find the fineness modulus and the zone each aggregate belongs to as per IS 383.

Sieve	Sample A retained (g)	Sample B retained (g)
10 mm	–	–
4.75 mm	5.9	1.4
2.36 mm	13.3	61
1.18 mm	115.5	114.2
600 µm	59.8	66.5
300 µm	90.7	67.8
150 µm	136.4	118.2
Pan	78.4	70.9

Total mass of each sample = **500 g**.

Sample A

Sieve	Retained (g)	% retained	Cumulative % retained	% passing
10 mm	0	0	0	100
4.75 mm	5.9	1.18	1.18	98.82
2.36 mm	13.3	2.66	3.84	96.16
1.18 mm	115.5	23.10	26.94	73.06
600 µm	59.8	11.96	38.90	**61.10**
300 µm	90.7	18.14	57.04	42.96
150 µm	136.4	27.28	84.32	15.68
Pan	78.4	15.68	–	–
			Σ = 212.22	

FM = 212.22 / 100 = 2.12 (fine sand).

Sample B

Sieve	Retained (g)	% retained	Cumulative % retained	% passing
10 mm	0	0	0	100
4.75 mm	1.4	0.28	0.28	99.72
2.36 mm	61.0	12.20	12.48	87.52
1.18 mm	114.2	22.84	35.32	64.68
600 µm	66.5	13.30	48.62	**51.38**
300 µm	67.8	13.56	62.18	37.82
150 µm	118.2	23.64	85.82	14.18
Pan	70.9	14.18	–	–
			Σ = 244.70	

FM = 244.70 / 100 = 2.45 (fine to medium sand).

Grading zone (IS 383 Table 9; the **600 µm sieve decides the zone**):

% passing 600 µm	Zone I 15–34	Zone II 35–59	Zone III 60–79	Zone IV 80–100
Sample A: 61.1			✓	
Sample B: 51.4		✓		

- **Sample A → Zone III.** Check other sieves against Zone III: 1.18 mm (73.1 vs 75 min) is 1.9% out and 300 µm (43.0 vs 40 max) is 3.0% out, a total of 4.9%, which is within the **5% tolerance** IS 383 allows on sieves other than 600 µm. But 15.7% passes 150 µm (limit 10%; 20% is allowed only for crushed sand), so it has **excess fines** and is acceptable only if it is crushed (manufactured) sand.
- **Sample B → Zone II.** 4.75 mm to 600 µm are all within Zone II, but 300 µm (37.8 vs 30 max) and 150 µm (14.2 vs 10 max) show it is **finer than Zone II at the fine end**; it should be checked by trial mixes or blended with coarser sand.

Q19

2 marks

> List out all the tests conducted on aggregate.

Property	Test (IS 2386 / IS 383)
Particle size and shape	**Sieve analysis** (grading, FM); **flakiness and elongation index**; angularity number
Deleterious materials	Clay, silt and fine dust (wet sieving, decantation); **organic impurities** (colour test with 3% NaOH); light-weight pieces; chlorides and sulfates
Specific gravity, density, voids, absorption	**Specific gravity and water absorption**; **bulk density** and voids; **bulking of sand**; moisture content
Mechanical	**Aggregate crushing value**; **impact value**; **Los Angeles abrasion value**; 10% fines value
Soundness	**Soundness** with sodium or magnesium sulfate
Alkali reactivity	Mortar bar / chemical method for **alkali–aggregate reactivity**; petrographic examination